Data Mining Techniques Process Report. Group 10.

Toni Kenđel[1 [2797438]], Kenia Lopez Sotomayor[1 [2682153]], and Iñigo Auza de la Mora[1 [2824537]]

Vrije Universiteit Amsterdam, 1081 HV Amsterdam, The Netherlands

1 Schedule and Task Assignment

During this assignment every member worked on the same model, although each member prepared differently the dataset to be used for training. This gave us more opportunities to apply different approaches to prepare the data for training. We strongly shared the approaches followed individually and built upon the data and model that gave the highest score after testing on the test data provided by the organizer of the competition

Table 1: Planning to carry out the task.

Date	Task	Description
22-28 April	Find ideas in Kaggle	Every member of the team had the task of looking for similar studies and obtaining ideas.
29-05 May	Selection and implementation of the model. Data cleaning. Feature engineering.	Toni worked on Random forest. Kenia and Iñigo worked on LightGBM.
06-19 May	Predictions. Scoring improvement.	Each member worked on the features to achieve higher score.
20-23 May	Writing report	All of us worked on the report. Data preparation (Toni and Kenia), Algorithm implementation and Evaluation (Iñigo).

2 Reflection on the Team's Interactions

During the course, our team maintained excellent communication regarding the tasks assigned to each member. We regularly shared our findings and discussed potential solutions to any problems that arose. This collaborative approach allowed us to leverage each other's strengths and insights, ensuring a well-rounded and thorough approach to our assignment.

3 What we learned

During this project, we discovered a new tool called LightGBM for analyzing data, of which none of us was familiar with. The data we had was full of different features

to look at, making it hard to pick the most important ones for our analysis. At first, we tried using all the features, which gave us a decent score, especially since we were beginners working on this machine learning model. But the real challenge came when we had to figure out which features to focus on. To improve our score, we had to play around with the features a lot. We looked at what others had done in similar projects to get ideas, and that helped us come up with better ways to use the features. Every time we submitted our work, we saw our score get a little better. It was motivating to see our progress on the leaderboard, and it pushed us to keep trying new things to make our model even better. Looking back, this project was tough but also a great learning experience. We got better at understanding how to work with data and make predictions, which will help us in future projects.